

A characteristic-zero counterexample to Tachikawa's second conjecture

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1 Statement and conventions

Theorem 1.1. *There exist a finite-dimensional symmetric $\mathbb{Q}$ -algebra Γ and a finite-dimensional nonprojective left Γ -module M such that*

$$\mathrm{Ext}_{\Gamma}^i(M, M) = 0 \quad (i > 0).$$

All tensor products without a subscript are over $k = \mathbb{Q}$, and $D = \mathrm{Hom}_k(-, k)$. For a finite-dimensional algebra G , set $G^e = G \otimes G^{\mathrm{op}}$. A symmetric algebra has a nondegenerate associative symmetric pairing, equivalently $G \simeq DG$ as bimodules. A bimodule is *side-projective* if it is projective separately as a left and as a right module; this does not mean projective over G^e .

Complexes have differentials lowering degree. We use $P[a]_j = P_{j-a}$ with differential $(-1)^a d_P$, and $c_j P = \mathrm{coker}(P_{j+1} \rightarrow P_j)$. Full Hom means

$$\mathrm{Hom}_G^a(P, Q) = \prod_{j \in \mathbb{Z}} \mathrm{Hom}_G(P_j, Q_{j-a}), \quad (\delta f)_j = d_Q f_j - (-1)^a f_{j-1} d_P.$$

For a chain map $v : P \rightarrow Q$,

$$\mathrm{Cone}(v)_j = Q_j \oplus P_{j-1}, \quad d_{\mathrm{Cone}(v)} = \begin{pmatrix} d_Q & v \\ 0 & -d_P \end{pmatrix}.$$

The tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy$. Stable Hom is module Hom modulo factorizations through finite projectives. For symmetric G , the stable shift $[1]$ is inverse syzygy.

2 The rational starting algebra

Let C have basis $(e, f, x, y, z, u, v, t, j, n)$ and corners

$$eCe = \langle e, x, y, z \rangle, \quad eCf = \langle u, v \rangle, \quad fCe = \langle t, j \rangle, \quad fCf = \langle f, n \rangle.$$

The idempotents are orthogonal, $1 = e + f$, and act according to these corners. The nonzero products of radical basis vectors are exactly

$$\begin{aligned} xy &= 2z, & yx &= z, & xu &= v, & yu &= v, \\ ut &= y + 2x, & uj &= z, & un &= v, & vt &= 2z, \\ tx &= j, & ty &= 4j, & tu &= 3n, & nt &= 2j. \end{aligned} \tag{1}$$

All other products of radical basis vectors vanish.

Lemma 2.1. *This table defines an associative algebra. Its radical is $J_C = \langle x, y, z, u, v, t, j, n \rangle$, and $J_C^5 = 0$. The diagonal map*

$$\sigma_0(e, f, x, y, z, u, v, t, j, n) = (e, f, -x, -y, z, u, -v, -t, j, -n)$$

is an involutive algebra automorphism. Its trace on each of the four corners of C is zero.

Proof. Give u, t degree one; x, y, n degree two; v, j degree three; and z degree four. The table is homogeneous. For associativity, idempotents and incompatible corners reduce immediately, and radical triples of degree greater than four vanish. The remaining compatible triples are

abc	utu	tut	xut	yut	ntu	unt	txu	tyu	utx	uty	tun
$(ab)c = a(bc)$	$3v$	$6j$	$2z$	$2z$	0	$2z$	0	0	z	$4z$	0 .

The positive-degree ideal has fifth power zero and quotient k^2 , so it is the radical. Each identity in (1) is homogeneous also for the displayed signs, proving multiplicativity of σ_0 . The corner sign lists are $(+, -, -, +)$, $(+, -)$, $(-, +)$, and $(+, -)$; each sums to zero. The multiplication table also gives

$$\text{tr}(L_a R_f \sigma_0^{-1}) = a_e - a_e + a_f - a_f = 0 \quad (a \in C). \quad (2)$$

Indeed the only possibly nonzero diagonal entries are at u, v, f, n ; they are $a_e, -a_e, a_f, -a_f$, respectively. This identity works over any commutative coefficient ring, not just over $\mathbb{Q}$. $\square$

Let s be the one-dimensional left C -module supported at f . For all $i \in \mathbb{Z}$ put $\ell_i = x - (-2)^i y$. Right and left multiplication by a are denoted ρ_a and λ_a .

Lemma 2.2. *A projective resolution $R \rightarrow s$ is*

$$R_0 = Cf, \quad R_i = Ce \ (i \geq 1), \quad d_1 = \rho_u, \quad d_{i+2} = \rho_{\ell_i} \ (i \geq 0).$$

Moreover

$$H^a \text{Hom}_C(R, s) = \begin{cases} k & a = 0, \\ 0 & a \neq 0, \end{cases} \quad H^a \text{Hom}_C(R, C) = \begin{cases} Ds & a = 2, \\ 0 & a \neq 2, \end{cases}$$

where the second equality respects the right C -module structure.

Proof. On the ordered basis (e, x, y, z, t, j) of Ce the relevant maps are

$$\begin{aligned} \rho_u &= (u, v, v, 0, 3n, 0), \\ \rho_{\ell_i} &= (\ell_i, -2(-2)^i z, z, 0, (1 - 4(-2)^i)j, 0). \end{aligned}$$

For $i \geq 0$, $1 - 4(-2)^i \neq 0$. Thus

$$\ker \rho_u = \langle \ell_0, z, j \rangle, \quad \text{im } \rho_{\ell_i} = \langle \ell_i, z, j \rangle, \quad \ker \rho_{\ell_i} = \langle \ell_{i+1}, z, j \rangle.$$

Also $\text{im } \rho_u = \text{rad}(Cf)$. This proves exactness at every index, without extrapolating from a finite number of degrees. Since $es = 0$ and $fs = k$, the first cohomology assertion follows.

The dual complex starts as $fC \xrightarrow{\lambda_u} eC \xrightarrow{\lambda_{\ell_0}} eC \xrightarrow{\lambda_{\ell_1}} \dots$. On (e, x, y, z, u, v) ,

$$\lambda_{\ell_i} = (\ell_i, -(-2)^i z, 2z, 0, (1 - (-2)^i)v, 0).$$

The map $\lambda_u : fC \rightarrow eC$ is injective, with image $\langle u, y + 2x, z, v \rangle$. At $i = 0$, the image of λ_{ℓ_0} is $\langle \ell_0, z \rangle$ and its kernel is $\langle \ell_{-1}, z, u, v \rangle$. For $i \geq 1$, its image and kernel are $\langle \ell_i, z, v \rangle$ and $\langle \ell_{i-1}, z, v \rangle$. Hence the sole cohomology is $k\bar{v}$ in degree two. Right multiplication by f fixes $\bar{v}$; the radical kills it, since $vt = 2z$ is a boundary. This is Ds . $\square$

Lemma 2.3. *The module DC has projective dimension at most two on both sides. Set*

$$T = C \ltimes DC, \quad B = C \otimes C, \quad E = T \otimes T.$$

Then T, E are symmetric, and

$$\mathrm{pd}({}_B E), \mathrm{pd}(E_B), \mathrm{id}({}_B B) \leq 4.$$

Proof. The coefficient of z in $eC \times Ce$ is a nondegenerate pairing: in the bases used above its determinant is -4 . Consequently $D(eC) \simeq Ce$ and $D(Ce) \simeq eC$. For the other summands there are exact sequences

$$\begin{aligned} 0 \rightarrow Cf &\xrightarrow{\rho_t} Ce \xrightarrow{\rho_{\ell-2}} Ce \xrightarrow{e \mapsto j^*} D(fC) \rightarrow 0, \\ 0 \rightarrow fC &\xrightarrow{\lambda_u} eC \xrightarrow{\lambda_{\ell_0}} eC \xrightarrow{e \mapsto v^*} D(Cf) \rightarrow 0. \end{aligned}$$

In each sequence the three nontrivial maps have ranks $4, 2, 4$. For the first, the last map sends (e, x, y, z, t, j) to $(j^*, t^*, 4t^*, 0, 2n^*, f^*)$; its kernel is $\langle \ell_{-2}, z \rangle$. The kernel of the middle map is $\langle y + 2x, z, t, j \rangle$, which is the image of $(f, u, v, n) \mapsto (t, y + 2x, 2z, 2j)$. For the second, the last map sends (e, x, y, z, u, v) to $(v^*, u^*, u^*, 0, n^*, f^*)$; its kernel is $\langle \ell_0, z \rangle$. The kernel of the middle map is $\langle y + 2x, z, u, v \rangle$, the image of $(f, t, j, n) \mapsto (u, y + 2x, z, v)$.

Alternatively, the right projective-dimension bound follows from an anti-involution. Define the rational linear map $\kappa : C \rightarrow C$ by

$$\kappa(x) = \tfrac{1}{2}y, \quad \kappa(y) = 2x, \quad \kappa(u) = t, \quad \kappa(t) = u, \quad \kappa(v) = 2j, \quad \kappa(j) = \tfrac{1}{2}v,$$

and fix e, f, z, n . The multiplication table directly gives $\kappa(ab) = \kappa(b)\kappa(a)$ and $\kappa^2 = 1$, hence $C \simeq C^{\mathrm{op}}$. The map $DC \rightarrow DC$, $\phi \mapsto \phi \circ \kappa$, is semilinear from the standard left dual action to the standard right dual action. It therefore preserves projectivity and the projective-dimension bound. A single left resolution of the whole dual is

$$0 \longrightarrow Cf \xrightarrow{\rho_t} Ce \xrightarrow{a \mapsto (0, a\ell_{-2})} Ce \oplus Ce \xrightarrow{(a, b) \mapsto a \cdot z^* + b \cdot j^*} DC \longrightarrow 0.$$

Its exactness follows from the displayed coordinate formulas.

Tensor the length-two resolutions over k to obtain the asserted projective-dimension bounds over B . Dualizing a right projective resolution of DB gives the injective bound. A trivial extension has symmetrizing functional $(a, \phi) \mapsto \phi(1)$, since its product pairing is $\phi(b) + \psi(a)$. The tensor product of two such forms proves symmetry of E . $\square$

3 Stable groups and the two scaling twists

Put $X = s \otimes s$, with E -action through $E \rightarrow B$, and let $\chi : E \rightarrow k$ be its character. We recall two standard facts about finite modules over a symmetric algebra G . Complete resolutions exist, and give

$$\underline{\mathrm{Hom}}_G(L, N[a]) \simeq H^a \mathrm{Hom}_G(P_L, N).$$

Tate duality gives natural composition pairings

$$D\underline{\mathrm{Hom}}_G(L, N) \simeq \underline{\mathrm{Hom}}_G(N, L[-1]). \quad (3)$$

One way to obtain this is to apply $D \mathrm{Hom}_G(P_j, N) \simeq \mathrm{Hom}_G(N, P_j)$ termwise to a complete resolution; the degree-zero homology on the dual side consists of maps into $\ker(P_0 \rightarrow P_{-1}) = \Omega L$, modulo projective factorizations. Naturality identifies the pairing with composition followed by a functional on $\underline{\mathrm{Hom}}_G(L, L[-1])$.

Proposition 3.1. *There is a class $\tau \in \text{Ext}_T^3(s, s)$ such that $\text{Ext}_T^*(s, s) = k[\tau]$ as a graded associative algebra, with $|\tau| = 3$. For $\mathcal{H}^a = \underline{\text{Hom}}_E(X, X[a])$, let*

$$P = k\langle \tau_1, \tau_2 \rangle / (\tau_1 \tau_2 + \tau_2 \tau_1), \quad \mathcal{P}_m = \langle \tau_1^a \tau_2^b : a + b = m \rangle.$$

Then X is nonprojective, $\mathcal{H}^{\geq 0} = P$, and the only nonzero groups are

$$\mathcal{H}^{3m} = \mathcal{P}_m, \quad \mathcal{H}^{-3m-1} = D\mathcal{P}_m \quad (m \geq 0). \quad (4)$$

Proof. Let $\mathcal{J} = T \otimes_C R$. The coefficient sequence $0 \rightarrow DC \rightarrow T \rightarrow C \rightarrow 0$ gives a triangle

$$s[2] \longrightarrow \mathcal{J} \longrightarrow s \xrightarrow{\tau} s[3]. \quad (5)$$

Indeed, for finite projective V , $DC \otimes_C V \simeq D \text{Hom}_C(V, C)$, so the first complex has just the homology s in degree two by Lemma 2.2. The complex $\mathcal{J}$ is bounded below projective, with bounded homology. Adjunction computes $\text{Hom}_{D^b(T)}(\mathcal{J}, s[i]) = k$ for $i = 0$ and zero otherwise. The long exact Hom sequence of (5) shows that composition with τ identifies $\text{Ext}_T^{i-3}(s, s)$ with $\text{Ext}_T^i(s, s)$ for every $i \geq 2$, and that $\text{Ext}_T^1(s, s) = 0$. Starting with $\text{End}_T(s) = k$ proves the first assertion, including products. No graded commutativity of this self-Ext algebra is asserted.

Tensor two projective resolutions of s with finite-dimensional terms. The Kunneth isomorphism and the usual signed external Yoneda product identify $\text{Ext}_E^*(X, X)$ with the graded tensor product of the two copies of $k[\tau]$. Since $3 \cdot 3$ is odd, the external generators anticommute. They have the PBW basis displayed in the statement. In particular degree three is nonzero, so X is not projective. Its ordinary endomorphism space is k ; its identity cannot factor through a projective, so the stable space is also k . Formula (3) now gives all negative groups. Under the composition pairing, postcomposition by τ_i on the negative groups is transpose to right multiplication by τ_i on P , with the shifts understood. In particular the map $\mathcal{H}^{-1} \rightarrow \mathcal{H}^2$ is zero. More explicitly, normalize the negative dual basis ϵ_{ab} by pairing $p \in \mathcal{P}_m$ with $\xi \in \mathcal{H}^{-3m-1}$ through the composite $p[-3m-1]\xi \in \mathcal{H}^{-1}$. With $a + b = m \geq 1$, this gives

$$\tau_1 \epsilon_{ab} = (-1)^b \epsilon_{a-1, b}, \quad \tau_2 \epsilon_{ab} = \epsilon_{a, b-1}, \quad (6)$$

where a negative index means zero. Indeed these operators are transpose to $p \mapsto p\tau_1$ and $p \mapsto p\tau_2$, respectively. Associativity of the displayed pairing proves this assertion without moving factors cyclically or suppressing a graded sign. When $m = 0$, both operators have the zero target $\mathcal{H}^2$. $\square$

For $H \in k^\times$, define $h_H(a, \phi) = (a, H\phi)$ on T , $\alpha_H = h_H \otimes h_H$ on E , and $U_H = E_{\alpha_H}$. The subscript twists the right action. The functor $\Phi_H = U_H \otimes_E -$ is restriction along α_H^{-1} : $a \otimes v \mapsto \alpha_H^{-1}(a)v$. It fixes X , using its character.

Lemma 3.2. *The induced actions on (4) are*

$$\Phi_H|_{\mathcal{H}^{3m}} = H^{-m} \text{id}, \quad \Phi_H|_{\mathcal{H}^{-3m-1}} = H^{m+2} \text{id} \quad (m \geq 0).$$

Proof. Twisting the coefficient sequence for $\mathcal{J}$ and comparing by $h_H \otimes 1$ acts by H on its $s[2]$ term and by 1 on its s term. Naturality gives $H\Phi_H(\tau) = \tau$. The positive formula follows by products.

Set $p = f \otimes f$ and $\zeta = f^* \otimes f^* \in Ep$. Multiplication on either side of ζ is by χ . The map $X \rightarrow \Omega X = (\text{rad } E)p$, $1 \mapsto \zeta$, is nonzero stably: symmetry implies that a vector annihilated on the left by $\text{rad } E$ is also annihilated on the right. A map $X \rightarrow E$ followed by a map $E \rightarrow (\text{rad } E)p$ is therefore zero, as are factorizations through finite sums and summands. The displayed nonzero map spans $\mathcal{H}^{-1}$. The comparison of the twisted projective cover $Ep \rightarrow X$ with itself is α_H , and sends ζ to $H^2\zeta$. The perfect composition pairing $\mathcal{H}^{3m} \times \mathcal{H}^{-3m-1} \rightarrow \mathcal{H}^{-1}$ then implies the negative weight: H^{-m} times the unknown weight must equal H^2 . $\square$

4 The twisted trace obstruction

Extend σ_0 to T by

$$\sigma(a, \phi) = (\sigma_0(a), \phi \circ \sigma_0^{-1}), \quad \theta = \sigma \otimes \sigma, \quad \theta_0 = \sigma_0 \otimes \sigma_0.$$

These are involutions, preserve the characters and vertex idempotents, and $\theta|_B = \theta_0$.

Lemma 4.1 (Reduction of a perfect factorization). *Let R be a finite-dimensional algebra of finite left self-injective dimension, and let ϑ be an automorphism. If a bimodule map $v : R_\vartheta \rightarrow DR$ factors in $D^b(R^e)$ through a perfect complex, it factors by ordinary maps through an object of $\text{add}(DR \otimes R)$, with the outer bimodule structure.*

Proof. For a bimodule L use the functorial left coinduction $J(L) = \text{Hom}_k(R, L)$, with $(a\phi b)(c) = \phi(ca)b$ and $L \hookrightarrow J(L)$ given by $z \mapsto (c \mapsto cz)$. Iterate its cokernel d times, where $d \geq \text{id}_{(R)R}$. For $L = R \otimes R$, this is the ordinary left injective coresolution of R , tensored with the right regular module; its last cokernel is injective by dimension shifting. Its terms therefore belong to $\text{add}(DR \otimes R)$. The construction preserves finite sums and split summands, so the same holds for every finite projective bimodule. Applying this finite functorial coresolution termwise to a bounded projective factorization complex and totalizing gives a bounded quasi-isomorphic complex P' with terms in $\text{add}(DR \otimes R)$.

For such a term V , both $\text{Ext}_{R^e}^i(R_\vartheta, V)$ and $\text{Ext}_{R^e}^i(V, DR)$ vanish for $i > 0$. For the first assertion, coinduction identifies the Hom complex with $\text{Hom}_{R^{\text{op}}}(\text{Res } R_\vartheta, R)$ computed from a restricted projective resolution; $(R_\vartheta)_R$ is free of rank one. For the second, a left projective resolution of DR tensored with the right regular module computes the group as $\text{Ext}_R^i(DR, DR) = 0$. Using a finite filtration by the terms of P' , these two vanishings show that the ordinary Hom complexes compute both derived arrows $R_\vartheta[0] \rightarrow P'$ and $P' \rightarrow DR[0]$. They consequently have actual chain-map representatives. Their composite factors through P'_0 , and equals v as a module map because degree-zero derived Hom between modules is ordinary Hom. $\square$

Proposition 4.2. *Write $\chi_B = \chi|_B$. The bimodule map*

$$\eta : B_{\theta_0} \longrightarrow DB, \quad \eta(a) = \chi_B(a)\chi_B,$$

does not factor through a perfect B^e -complex.

Proof. It is a bimodule map since $\chi_B\theta_0 = \chi_B$. Lemma 2.3 permits Lemma 4.1 to be applied. Consider an ordinary factorization $B_{\theta_0} \xrightarrow{\alpha} DB \otimes B \xrightarrow{\beta} DB$. Identify $DB \otimes B$ with $\text{Hom}_k(B, B)$ by $\lambda \otimes c \mapsto (x \mapsto \lambda(x)c)$. The vector $\alpha(1)$ corresponds to a map F satisfying

$$F(x\theta_0(a)) = F(x)a, \quad \text{hence} \quad F(x) = d\theta_0^{-1}(x)$$

for some $d \in B$. If (c_i) is a basis with dual (c_i^*) , this says $\alpha(1) = \sum_i c_i^* \otimes d\theta_0^{-1}(c_i)$. Every possible β has the form $\beta(\lambda \otimes c) = \lambda bc$ for some $b \in B$: its restriction at $c = 1$ is a left-module endomorphism of DB , dual to left multiplication on the right regular module. Since $\theta_0(p) = p$, evaluation gives

$$(\beta\alpha(1))(p) = \sum_i c_i^*(bd\theta_0^{-1}(c_i)p) = \text{tr}(L_{bd}R_p\theta_0^{-1}). \quad (7)$$

For a pure tensor $a \otimes c \in B$, the operator in (7) is $(L_a R_f \sigma_0^{-1}) \otimes (L_c R_f \sigma_0^{-1})$. The trace of a tensor product is the product of traces, so (2) makes it zero. Pure tensors span B ; linearity gives the same conclusion for bd . This direct computation avoids a radical-filtration argument entirely. The conclusion persists for finite sums and summands of $DB \otimes B$. But $\eta(1)(p) = 1$, which is a contradiction. $\square$

5 A finite bimodule and its evaluated Hom profile

Let K be the two-sided bar model of $E \otimes_B^L E$: $K_i = E \otimes B^{\otimes i} \otimes E$ for $i \geq 0$. The differential is the sum of the consecutive multiplication maps with signs $(-1)^r$, $0 \leq r \leq i$. Write K_θ for its right twist. Its degree-zero multiplication map has target E_θ .

We use the stable image of a bounded derived object over a symmetric algebra: if P is a bounded-below finite projective model with bounded homology, then $\text{st}(P) = (c_n P)[n]$ for any sufficiently large n . The short exact syzygy sequences make this independent of n , and comparison maps make it functorial. Equivalently this is the standard identification of the singularity category with the stable module category. A derived map with zero stable image factors through a perfect complex. Indeed its map on sufficiently high cokernels factors through a projective; lifting that factorization and successively extending a homotopy removes the high components, leaving a factorization through a bounded projective truncation. We will use this statement over E^e , which is symmetric too.

Lemma 5.1. *Set $\mathcal{I} = \text{st}(K_\theta)$. It has a finite side-projective representative. So does $Y = \mathcal{I}[-1]$. Evaluation on X gives*

$$YX \simeq \text{st}(\mathcal{J} \otimes \mathcal{J})[-1].$$

Here and below YX abbreviates $Y \otimes_E X$ in the stable category.

Proof. On either side the untwisted bar model is homotopy equivalent to the induction of a projective B -resolution of E of length at most four. This follows from Lemma 2.3 and comparison of ordinary projective resolutions; tensoring preserves that homotopy equivalence. Twisting does not change these bounds. Hence K_θ has no homology above degree four, and $c_5 K_\theta$ is projective on both sides. More explicitly a side homotopy to a complex ending in degree four gives $1 = d_6 h_5 + h_4 d_5$ in degree five; modulo $\text{im } d_6$, the identity on $c_5 K_\theta$ factors through K_4 , making it a projective summand. Take four finite E^e cosyzygies of $c_5 K_\theta$ to represent Y . They remain side-projective, since embeddings into finite bimodule projective-injectives split on each side.

Evaluation on X sends bimodule projectives to E -projectives and preserves these side-split exact sequences, hence commutes with these stable images and shifts. Since θ fixes the character of X , $K_\theta \otimes_E X \simeq K \otimes_E X$. The latter is the induction of the B -bar resolution of X . Comparison with $R \otimes R$ gives its homotopy equivalence to $\mathcal{J} \otimes \mathcal{J}$. $\square$

Lemma 5.2 (The signed Koszul calculation). *The bimodule in Lemma 5.1 satisfies*

$$W^a := \underline{\text{Hom}}_E(X, YX[a]) = \begin{cases} k & a = -3, 0, \\ 0 & \text{otherwise.} \end{cases} \quad (8)$$

Proof. Choose a projective model of (5) with underlying graded module $A \oplus Q$, where A resolves $s[2]$, Q resolves s , and the off-diagonal differential represents τ . Tensor two copies and shift by $[-4]$. The resulting filtered object $\mathcal{L}$ has three levels

$$X, \quad X[-2]^2, \quad X[-4].$$

The attaching row and column can be identified with (τ_1, τ_2) and $(\tau_2, \tau_1)^t$. These signs come from the tensor differential: the off-diagonal maps in different factors are odd, and their compositions cancel. The shifts identifying the cells differ by two, so introduce no additional interchange sign. Equivalently their composite is $\tau_1 \tau_2 + \tau_2 \tau_1 = 0$, not a relation in a commutative polynomial ring.

For each integer l the induced adjacent maps are

$$d_l : (\mathcal{H}^{l-2})^2 \rightarrow \mathcal{H}^{l+1}, \quad (u, v) \mapsto \tau_1 u + \tau_2 v, \quad d'_l : \mathcal{H}^{l-4} \rightarrow (\mathcal{H}^{l-1})^2, \quad w \mapsto (\tau_2 w, \tau_1 w).$$

Composition here is shifted postcomposition. On the positive groups the exact algebraic sequence is

$$0 \rightarrow \mathcal{P}_{m-1} \xrightarrow{(\tau_2, \tau_1)^t} \mathcal{P}_m^2 \xrightarrow{(\tau_1, \tau_2)} \mathcal{P}_{m+1} \rightarrow 0 \quad (m \geq 0), \quad (9)$$

where $\mathcal{P}_{-1} = 0$. The row is onto, the column is injective, and their composite is zero. The PBW basis gives dimensions $m, 2m+2, m+2$, so the middle is exact. The right-multiplication version is exact by the same reasoning. On the negative groups use its dual and interchange the two middle slots, as follows from the composition pairing. For completeness, positive exactness has an explicit coordinate certificate. If u_i, v_i are the coefficients of the two degree- m inputs in the basis $\tau_1^i \tau_2^{m-i}$, the row has coefficients $u_{i-1} + (-1)^i v_i$, with missing coefficients set to zero. A section sends z to $u_i = z_{i+1}$ and $v_i = \delta_{i0} z_0$. For a vector in the kernel, a column preimage is $w_i = (-1)^i u_i$ for $0 \leq i < m$; the last row coefficient gives $u_m = 0$. The right-multiplication sequence is proved in the same way; its first operator has the factor $(-1)^{m-i}$ on the i th monomial. Its dual gives the negative sequence with the two middle slots exchanged. Concretely, (6) makes the negative row $(D\mathcal{P}_m)^2 \rightarrow D\mathcal{P}_{m-1}$ surjective and the negative column $D\mathcal{P}_{m+1} \rightarrow (D\mathcal{P}_m)^2$ injective, with zero composite, for $m \geq 1$. Their dimensions are $m, 2m+2$, and $m+2$, so the middle is exact. At the seam the map $D\mathcal{P}_1 \rightarrow (D\mathcal{P}_0)^2$ is an isomorphism; the next target is the actual zero group $\mathcal{H}^2$. Consequently, in every degree,

$$\begin{aligned} \ker d_l &= \text{im } d'_{l-1}, \\ \text{coker } d_{l-1} &= \begin{cases} k & l = 0, \\ 0 & l \neq 0, \end{cases} \quad \ker d'_l = \begin{cases} k & l = 3, \\ 0 & l \neq 3. \end{cases} \end{aligned}$$

It is important to retain the full top attaching map. Let $\mathcal{L}_1$ be the bottom two levels and $N^l = \underline{\text{Hom}}_E(X, \mathcal{L}_1[l])$. The first long exact sequence gives $0 \rightarrow \text{coker } d_{l-1} \rightarrow N^l \xrightarrow{q_l} \ker d_l \rightarrow 0$. Write $r_l : \mathcal{H}^{l-4} \rightarrow N^{l+1}$ for the actual top connecting map. It satisfies $q_{l+1} r_l = d'_l$. Thus $\ker r_l = 0$ except possibly at $l = 3$; there $N^4 = 0$, so the kernel is $\mathcal{H}^{-1} = k$. For $l \neq 0$, q_l is an isomorphism and r_{l-1} is onto. At $l = 0$, $N^0 = k$ and the incoming source $\mathcal{H}^{-5}$ is zero. The final long exact sequence therefore leaves precisely k in degrees $l = 0, 3$ for $\underline{\text{Hom}}_E(X, \mathcal{L}[l])$. Since $YX = \mathcal{L}[3]$, this is (8). This argument also covers any additional component of the top attaching map into the bottom level; no splitting of the full filtration is assumed. $\square$

6 Obtaining genuine bimodule lifts

Let $S = \text{Hom}_k(X, X)$ with $(atb)(x) = a t(bx)$. Both actions on its one-dimensional underlying space are given by χ .

Lemma 6.1 (Stable adjunction). *For a finite side-projective E -bimodule L , adjunction descends to $\underline{\text{Hom}}_{E^e}(L, S) \simeq \underline{\text{Hom}}_E(LX, X)$.*

Proof. A bimodule projective factorization evaluates to a projective module factorization. Conversely embed L in a finite E^e -projective-injective J . This is split as a right E -map, so $LX \hookrightarrow JX$. If the evaluated map factors through a finite projective E -module P , injectivity of P extends its first arrow from LX to JX . Ordinary adjunction then gives a factorization of the original map through J . The factorization ideals coincide. $\square$

Proposition 6.2. *For every $H \in k^\times$, there is an ordinary bimodule map $g_H : U_H \rightarrow Y$ whose evaluation represents a nonzero element of W^0 .*

Proof. The twisted multiplication followed by the character gives a derived map $K_\theta \rightarrow E_\theta \rightarrow S$, with stable image $\phi : \mathcal{I} \rightarrow S$. The map $b_0 : S \rightarrow U_H$, $1 \mapsto \zeta$, is a bimodule map because $\chi \alpha_H = \chi$. We claim $b_0 \phi \neq 0$ in the stable category of E^e . Otherwise its derived

representative factors through a perfect E^e complex. Restricting to B^e still leaves a perfect complex: a free E^e -module restricts to $E \otimes E$, whose projective dimension over B^e is at most eight by Lemma 2.3.

Precompose with the inclusion of the right- θ_0 -twisted bar resolution of B into $\text{Res } K_\theta$, and postcompose with $\text{Res } U_H \rightarrow DB$ from

$$\text{Res } U_H = B \oplus (DC \otimes C) \oplus (C \otimes DC) \oplus DB.$$

This is a decomposition of B -bimodules since α_H fixes B . In degree zero the composite sends $a \otimes c$ to $\chi_B(ac)\chi_B$, and it is zero in positive degrees. It is exactly η composed with the twisted bar augmentation. We would have obtained a perfect factorization of η , contradicting Proposition 4.2.

Apply symmetric stable duality over E^e to

$$\phi_* : \underline{\text{Hom}}_{E^e}(U_H[1], \mathcal{I}) \longrightarrow \underline{\text{Hom}}_{E^e}(U_H[1], S).$$

Its transpose is $b \mapsto b\phi$ from $\underline{\text{Hom}}_{E^e}(S, U_H)$ to $\underline{\text{Hom}}_{E^e}(\mathcal{I}, U_H)$. It is nonzero by b_0 , so there exists $\beta : U_H[1] \rightarrow \mathcal{I}$ with $\phi\beta \neq 0$. Stable adjunction applies to the side-projective source $U_H[1]$. It identifies this nonzero class with the composite of $\beta X : X[1] \rightarrow \mathcal{I}X$ and the stable adjoint $\mathcal{I}X \rightarrow X$ of ϕ . Thus $\beta X \neq 0$. Shift by $[-1]$, use $Y = \mathcal{I}[-1]$, and choose an ordinary representative of the resulting stable bimodule map. Its evaluation is nonzero, as required. No evaluation functor is applied to arbitrary shifts of the nonside-projective bimodule S . $\square$

7 Two branches and the comparison maps

Take $H_1 = 2$, $H_2 = 3$, and normalize the lifts so that $g_1X = g_2X = w \neq 0$ in W^0 . Choose a finite free bimodule Q surjecting onto Y and form

$$0 \rightarrow F \rightarrow U_2 \oplus U_3 \oplus Q \xrightarrow{(g_1, -g_2, \pi)} Y \rightarrow 0. \quad (10)$$

The minus sign is essential for the next normalization. The sequence splits on each side, so F is finite and side-projective. Put $\mathcal{V}^a = \underline{\text{Hom}}_E(X, FX[a])$. Evaluation and stable Hom give

$$(\mathcal{H}^{a-1})^2 \rightarrow W^{a-1} \rightarrow \mathcal{V}^a \xrightarrow{p^a} (\mathcal{H}^a)^2 \rightarrow W^a.$$

In degree zero the last arrow is $(c, d) \mapsto (c - d)w$. There is a unique $v \in \mathcal{V}^0$ with $p^0(v) = (1, 1)$. Define

$$\Delta^a : (\mathcal{H}^a)^2 \rightarrow \mathcal{V}^a, \quad (g, h) \mapsto F(g)v - v[a]h.$$

Naturality of the projections onto the two twists gives

$$p^a \Delta^a(g, h) = (\Phi_2(g) - h, \Phi_3(g) - h). \quad (11)$$

Lemma 7.1. *The map Δ^0 is onto. The map Δ^a is invertible for $a > 0$, $a \leq -3$, and $a = -1$. The map $\Delta^{-2} : 0 \rightarrow k$ is injective.*

Proof. Using (8), p^a is an isomorphism for $a > 0$, $a \leq -3$, and $a = -1$. At $a = 1$ this also uses surjectivity of the degree-zero map to W^0 ; at $a = -3$ source and target are both zero. The remaining values are $\mathcal{V}^{-2} = k$ and $\mathcal{V}^0 = kv$. On a supported positive degree $3m$, (11) is

$$\begin{pmatrix} 2^{-m} & -1 \\ 3^{-m} & -1 \end{pmatrix} \otimes \text{id}_{\mathcal{P}_m} \quad (m \geq 1),$$

with determinant $3^{-m} - 2^{-m} \neq 0$. On a supported negative degree $-3m - 1$ it is

$$\begin{pmatrix} 2^{m+2} & -1 \\ 3^{m+2} & -1 \end{pmatrix} \otimes \text{id}_{D\mathcal{P}_m} \quad (m \geq 0),$$

with determinant $3^{m+2} - 2^{m+2} \neq 0$. At $m = 0$ this determinant is 5. Unsupported groups vanish. Finally $\Delta^0(c, d) = (c - d)v$ is onto, and $\mathcal{H}^{-2} = 0$ gives the stated map in degree -2 . $\square$

8 The triangular complete resolution

Set $\Lambda = \begin{pmatrix} E & 0 \\ F & E \end{pmatrix}$. Write its modules as triples $(L, N, \alpha : FL \rightarrow N)$ and use the columns $D_0(L) = (L, FL, 1)$ and $D_1(L) = (0, L, 0)$. They are exact and preserve projectives. Their Hom spaces satisfy

$$\begin{aligned} \text{Hom}(D_i L, D_i L') &= \text{Hom}_E(L, L') \quad (i = 0, 1), \\ \text{Hom}(D_0 L, D_1 L') &= 0, \\ \text{Hom}(D_1 L, D_0 L') &= \text{Hom}_E(L, FL'). \end{aligned}$$

Choose a complete resolution P of X , with $P_{-1} = Ep$ and $d_0 = i_X \epsilon$, where $i_X(1) = \zeta$ and $\epsilon : P_0 \rightarrow X$. An ordinary representative $\tilde{v} : X \rightarrow FX$ lifts to a chain map $P \rightarrow FP$. For completeness, positive-degree lifts use projectivity and exactness; downward lifts use exactness of $\text{Hom}_E(P, J)$ for projective J . Thus there is a map $V : D_1(P) \rightarrow D_0(P)$. Put $P_Z = \text{Cone}(V)$ and $Z = c_0 P_Z$.

Proposition 8.1. *The complex P_Z is totally acyclic, Z is finite and nonprojective, and*

$$H^a \text{Hom}_\Lambda(P_Z, Z) = \begin{cases} k & a = -1, 0, \\ 0 & a \notin \{-1, 0\}. \end{cases} \quad (12)$$

Proof. The column complexes are exact and projective. Their Hom complexes into $\Lambda = D_0(E) \oplus D_1(E)$ are sums of $\text{Hom}_E(P, E)$, $\text{Hom}_E(P, F)$, and zero. They are exact, since F is finite projective on the left. Hence their cone is totally acyclic. Explicitly Z is the triple with upper component X and lower component

$$(FX \oplus Ep) / \{(\tilde{v}(x), -i_X(x)) : x \in X\}.$$

This follows by first quotienting the FP_0 component by FP_1 and then using $d_0 = i_X \epsilon$ in the cone relation. All terms are finite. The upper-component functor sends finite Λ -projectives to E -projectives. Since X is nonprojective, so is Z .

Let $A_0 = D_0(P)$, $B_0 = D_1(P)$. The zero cross block $\text{Hom}^\bullet(A_0, B_0[1])$ gives a short exact sequence of full Hom complexes with diagonal quotient $\text{End}^\bullet(A_0) \oplus \text{End}^\bullet(B_0[1])$ and kernel $\text{Hom}^\bullet(B_0[1], A_0)$. Its cohomology in degree a is $\mathcal{V}^{a-1}$. The connecting map on the diagonal degree- a cohomology is Δ^a , up to an overall nonzero sign. Here is an explicit sign check. If g, h are degree- a diagonal cocycles, identify the second diagonal on $B_0[1]$ by $h'_j = (-1)^a h_{j-1}$. The cross component of its differential is $(-1)^a (Vh - gV)$. After the standard shift identification this is precisely the difference defining Δ^a , up to a common sign. Thus kernels and cokernels are exactly those in Lemma 7.1.

It follows that

$$0 \rightarrow \text{coker } \Delta^{a-1} \rightarrow H^a \text{End}_\Lambda^\bullet(P_Z) \rightarrow \ker \Delta^a \rightarrow 0.$$

For $a > 1$ and $a \leq -3$ both ends vanish. At $a = 1$, use surjectivity of Δ^0 and injectivity of Δ^1 ; at $a = -2$, use surjectivity of Δ^{-3} and injectivity of Δ^{-2} . The other two degrees have dimensions one because Δ^{-1} is invertible, $\text{coker } \Delta^{-2} = k$, and $\ker \Delta^0 = k$.

Finally, complete-resolution comparison identifies $H^a \text{End}_\Lambda^\bullet(P_Z)$ with $H^a \text{Hom}_\Lambda(P_Z, Z)$. To see why it applies although Λ need not be self-injective, total acyclicity extends every map from a cokernel of P_Z to a projective across the adjacent projective term. This gives downward comparison lifts and the corresponding nullhomotopies. Taking cokernels therefore identifies full-chain-map homotopy classes with stable module maps, which are also the cycles modulo boundaries of $\text{Hom}(P_Z, Z)$. Full Hom uses products, so all components of these lifts are allowed. This proves (12). $\square$

9 Transfer and completion

Lemma 9.1. *Let R be a finite-dimensional algebra and let P be a totally acyclic complex of finite projectives with nonprojective $Z = c_0 P$. If $H^a \text{Hom}_R(P, Z) = 0$ for $a > 0$ and $a \leq -2$, then*

$$G = R \ltimes DR, \quad M = G \otimes_R Z$$

satisfy: G is symmetric, M is finite and nonprojective, and $\text{Ext}_G^i(M, M) = 0$ for every $i > 0$.

Proof. Let $I = DR \otimes_R P$. Termwise finite projectivity gives $I_j \simeq D \text{Hom}_R(P_j, R)$, so I is exact by total acyclicity and has injective terms. Restriction of $G \otimes_R P$ is $P \oplus I$, hence exact, and its terms are finite projective over G . Right exactness gives its degree-zero cokernel M ; its nonnegative tail is therefore a projective resolution of M . No flatness of G_R is asserted or used.

Set $N = DR \otimes_R Z = c_0 I$. The negative tail $0 \rightarrow N \rightarrow I_{-1} \rightarrow I_{-2} \rightarrow \cdots$ is an injective coresolution. The evaluation isomorphism $\text{Hom}_R(Z, I_j) \simeq D \text{Hom}_R(P_j, Z)$ gives

$$\text{Ext}_R^i(Z, N) \simeq DH^{-i-1} \text{Hom}_R(P, Z) = 0 \quad (i > 0).$$

In particular $i = 1$ uses degree -2 , not degree -1 . Induction–restriction adjunction, on the projective resolution already constructed, now yields

$$\text{Ext}_G^i(M, M) \simeq \text{Ext}_R^i(Z, Z \oplus N) = 0.$$

Symmetry of the trivial extension was proved in Lemma 2.3. If M were projective, applying $R \otimes_G -$ to a split free presentation would make $R \otimes_G M \simeq Z$ projective. A tensor functor preserves a split identity composite without a flatness assumption. This is a contradiction. $\square$

Proof of Theorem 1.1. Apply Lemma 9.1 to the finite-dimensional algebra $R = \Lambda$ and the complete resolution in Proposition 8.1. It gives $\Gamma = \Lambda \ltimes D\Lambda$ and $M = \Gamma \otimes_\Lambda Z$ with exactly the asserted properties. All the algebras and modules in the construction are defined over $\mathbb{Q}$. $\square$